

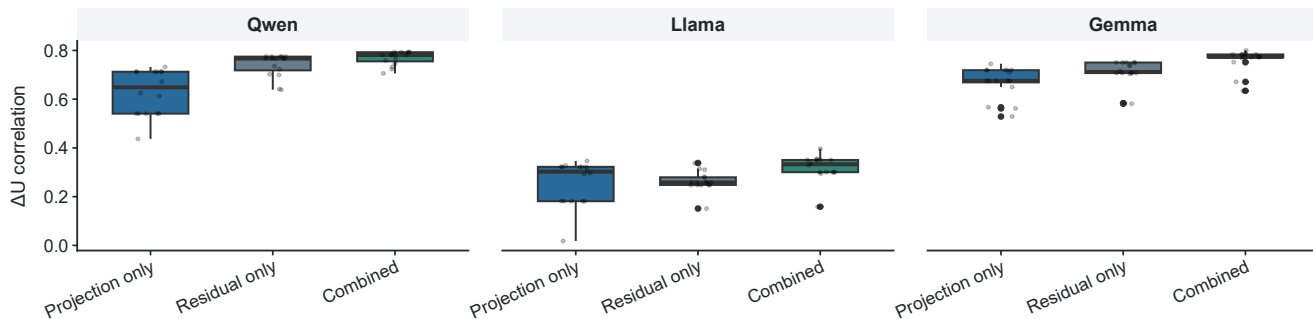
Residual trajectory features retain structured and complementary signal

a



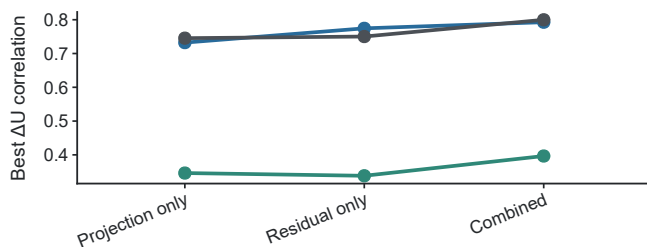
b Residual features retain ΔU information across windows

All evaluated windows and K values; grouped-CV correlations



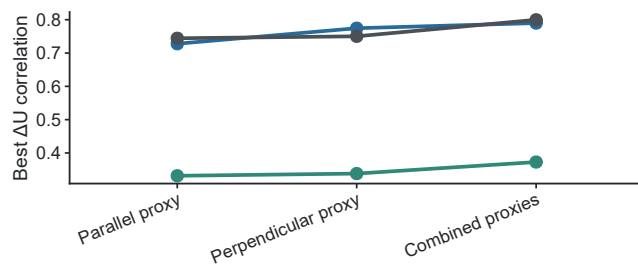
c Best-window ΔU summaries are checkpoint dependent

Qwen Llama Gemma



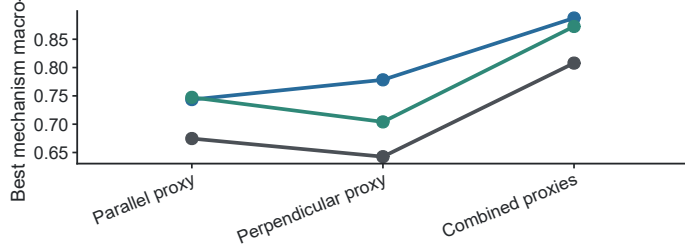
d Parallel and perpendicular proxy families are complementary

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e Combined proxies improve mechanism-label separation

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f

Retained conclusion

Structured residual

Proxy families are not an exhaustive decomposition